Part III (Standalone): Verification Program

A two-level audit protocol for the Two-Substrate Theory + Loss-Tracing Tool (LTT)

This document is a dedicated verification plan. It does not reprint Parts I or II.

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Abstract

This document specifies a verification program for the framework developed in Parts I and II: (1) the Two-Substrate + Void bookkeeping worldview and (2) the operational Loss-Tracing Tool (LTT). The program is organized into two levels: **computational validation** ("code experiment") and **real-physics validation** (laboratory and observational domains). A strict gating architecture prevents premature ontological claims: the LTT must pass well-definedness, calibration, closure, identifiability, and intervention tests before any dark-sector interpretations are discussed. The output of this program is a "tool certificate" (what the LTT can and cannot claim), plus domain-specific reports that enable falsifiable hypothesis comparison.

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1 Scope, Inputs, and What This Part Verifies

1.1 Inputs from Parts I and II

This verification program assumes the following objects already exist (as defined in Parts I and II):

- A **model contract** defining observables $\mathcal{O}$, projections Π_X, Π_T, calibration geometry $\mathbf{W}$ (or likelihood metric), channelization $\mathcal{C}$, and ledger update maps for V_E, V_I.
- A clear separation between:
  1. **Operational tool layer** (LTT as residual/mismatch pipeline),
  2. **Interpretive layer** ("Void as physical" and dark-sector claims as hypotheses).

1.2 What this Part verifies (and what it does not)

Verified here:

1. The LTT is *well-defined, computable, calibrated*, and *does not collapse into a garbage bin.*
2. The LTT can distinguish (within an explicit contract) between:
   - missing-channel loss,
   - systematic model error, and
   - stochastic noise.
3. The LTT can be deployed as a disciplined residual microscope across domains.

Not "proven" here: any ontological claim that "Void is a new physical substance" or that any specific residual is dark matter/energy. Those are hypotheses evaluated only after the tool passes all gates.

2 Verification Philosophy and the Gating Rule

Postulate 2.1 (Gating rule (no jumping layers))**.** No interpretation beyond the operational tool layer is allowed unless the tool passes all gates at Level 1 and at least one real-physics domain at Level 2.

Definition 2.2 (Tool certificate)**.** A **tool certificate** is a structured report stating:

- which contracts were tested,
- which gates were passed/failed,
- what residual structures can be reliably detected,
- what claims are *invalid* (e.g., "Void explains X") under the tested contracts.

2.1 Four primary gates

We define four non-negotiable gates:

1. **Well-definedness:** $\mathcal{O}, \Pi_X, \Pi_T, \mathbf{W}, \mathcal{C}$ and uncertainty models exist and are coherent.
2. **Closure:** in a closed accounting system, expected residual channels are near zero.
3. **Identifiability:** the tool separates missing-channel loss from model/systematic error.
4. **Intervention:** controlled changes produce predicted directional changes in δ and in channelized loss.

Two-Level Verification Flow

$$\underbrace{\text{Level 1: Code}}_{\text{tool correctness}} \longrightarrow \underbrace{\text{Tool Certificate}}_{\text{pass/fail gates}} \longrightarrow \underbrace{\text{Level 2: Real Physics}}_{\text{domain deployment}} \longrightarrow \underbrace{\text{Hypothesis Comparison}}_{\text{H0/H1/H2}}$$

Figure 1: The gating architecture prevents "interpretation first, audit later."

3 Level 1: Computational Validation (Code Experiments)

3.1 Level-1 deliverables

Level 1 must produce:

1. **LTT Core Library** (minimal API):
   - `define_contract()` $\Rightarrow \mathcal{O}, \Pi_X, \Pi_T, \mathbf{W}, \mathcal{C}, h_E, h_I$
   - `fit_baseline()` $\Rightarrow$ nuisance/systematics fit + uncertainty propagation
   - `compute_residual()` $\Rightarrow r, \delta, \mathbf{L}, V_E, V_I$
   - `structure_tests()` $\Rightarrow$ periodicity/coherence/lines/changepoints
   - `report_certificate()` $\Rightarrow$ gate pass/fail + limitations
2. **Synthetic Data Generators** for:
   - closed energy-budget systems,
   - missing-channel scenarios,
   - systematic model-error scenarios,
   - particle-like vs field-like event families.
3. **Automated Gate Suite** producing a pass/fail matrix and artifacts (plots, confusion matrices, ablation tables).

3.2 Level-1 gate tests (formal)

Definition 3.1 (Level-1 acceptance criterion)**.** A Level-1 gate is passed if it meets its quantitative threshold under cross-validation and sensitivity analysis (ablation).

3.2.1 Gate 1: Well-definedness (contract coherence)

Checklist:

- $\mathcal{O}$ components have units (or are dimensionless), calibration procedure, and uncertainty model.
- Π_X, Π_T output $\hat{y} \in \mathcal{O}$ of identical dimension and semantics.
- $\mathbf{W}$ is positive semidefinite and matches $\mathcal{O}$ ordering.
- $\mathcal{C}$ is explicitly defined and not post-hoc adjusted to hide failures.

3.2.2 Gate 2: Closure (closed accounting sandbox)

Build a sandbox with full channel instrumentation (ground truth). Require:

$$\mathbb{E}[\mathbf{L}] \approx \mathbf{0} \quad \text{within propagated uncertainty.}$$

Failure modes:

- miscalibration of Π,
- incorrect noise model $\mathbf{\Sigma}$,
- missing channel in $\mathcal{O}$,
- channelization mismatch $\mathcal{C}$.

3.2.3 Gate 3: Identifiability (loss vs model error)

Test two controlled conditions:

- **Missing-channel injection:** remove one sensor/channel; residual must move into the corresponding channel proxy.
- **Systematic model error:** deliberately bias Π_T while keeping full channels; tool must flag "model error" rather than crediting Void.

Require a confusion matrix with high separation between "missing-channel" and "systematic error" cases under cross-validation.

3.2.4 Gate 4: Intervention response

Introduce controlled parameter changes (e.g., friction coefficient, shielding level, temperature-dependent noise). Require:

- correct *directional* change in δ and in target channels,
- stable behavior under small perturbations (robustness).

3.3 Particle-like vs field-like discrimination (code level)

This implements the conceptual point: detector events can look particle-like even if driven by a coherent field.

Definition 3.2 (Particle-like vs field-like signatures (operational))**.** In code experiments:

- **Particle-like:** stochastic event times (Poisson-like), broad continuum spectra, weak cross-sensor coherence.
- **Field-like:** coherent time structure (periodic/quasi-periodic), narrow spectral lines, cross-channel and cross-detector coherence.

Algorithm 1: Level-1 Certification Pipeline (Automated Gates)

Input: Contract $(\mathcal{O}, \Pi_X, \Pi_T, \mathbf{W}, \mathcal{C})$; synthetic generators $\mathcal{G}_{\text{closed}}, \mathcal{G}_{\text{missing}}, \mathcal{G}_{\text{sys}}, \mathcal{G}_{\text{particle}}, \mathcal{G}_{\text{field}}$.

Output: Tool certificate + gate matrix + artifacts.

Init *Initialize calibration from closed-control data; freeze contract parameters.*
Run Gate-1 coherence checks; fail if contract ill-typed.
Run closed sandbox; compute $\mathbb{E}[\mathbf{L}]$ and uncertainties; decide Gate-2.
Run missing-channel and systematic-error suites; compute confusion matrix; decide Gate-3.
Run interventions; test directional predictions; decide Gate-4.
Run particle-vs-field suites; compute structure-test sensitivity/specificity.
Return certificate with explicit limitations (what cannot be claimed).

3.4 Exit criteria for Level 1

Level 1 passes when:

- all four gates pass under ablation and cross-validation,
- tool certificate is produced with reproducible artifacts,
- particle-vs-field discrimination is demonstrated *on synthetic truth.*

4 Level 2: Real-Physics Validation (Domains)

4.1 General domain protocol (applies to every domain)

Every domain must specify:

1. **Contract specification:** $\mathcal{O}, \Pi_X, \Pi_T, \mathbf{W}, \mathcal{C}$ (frozen before final analysis).
2. **Calibration source:** control data or instrument models defining $\mathbf{\Sigma}$ / likelihood geometry.
3. **Systematics list:** known nuisance effects; how they enter Π and $\mathbf{\Sigma}$.
4. **Intervention or perturbation:** what can be varied to test directional predictions.
5. **Gate mapping:** how closure/identifiability/intervention are operationally tested here.

Remark 4.1. Not every domain has physical interventions (cosmology cannot "turn knobs"). In that case, intervention is replaced by *systematics perturbation* (varying nuisance models, selection functions, calibration pipelines).

4.2 Domain 2.1: Laboratory closure (energy accounting)

Goal: prove LTT works on real signals with full accounting before any dark-sector analysis.

Example systems (choose at least one):

- mechanical oscillator with friction + thermal sensor,
- electrical RLC circuit with Joule heating + radiative leakage proxy,
- coupled oscillator network (complex but controllable).

Domain gates:

- **Closure:** measure input power and all major output channels; residual must vanish within uncertainty.
- **Identifiability:** intentionally remove one sensor and confirm residual moves into the right channel.
- **Intervention:** modify friction/load/shielding and predict channel shifts.

Pass condition: lab closure and interventions pass with reproducible results across multiple runs.

4.3 Domain 2.2: Rare-event / detector-style residual analysis

Goal: verify LTT can audit a complex background model without inventing "new physics" by default.

Protocol template:

- $\mathcal{O}$: binned spectra + discriminants + time bins (and spatial bins if available).
- Metric: Poisson deviance or likelihood ratio.
- $\mathcal{C}$: channels for ER/NR-like components, surface events, radiogenic backgrounds, time-dependent modulations.
- Systematics: detector response uncertainties, efficiency curves, background composition uncertainty.

Domain gates:

- **Closure surrogate:** fit on control regions; residual should be consistent with noise under baseline.
- **Identifiability:** inject synthetic signals (blinded) into real backgrounds and verify recovery.
- **Intervention surrogate:** compare runs with different operating conditions (or different analysis cuts) and require residual behavior consistent with known changes.

Pass condition: tool reproduces baseline results and detects only structured residuals that survive systematics perturbations.

4.4 Domain 2.3: Cosmology / astrophysical residual mapping

Goal: treat "dark sector" as operational residual mapping under explicit contracts, not as premature ontology.

Protocol template:

- $\mathcal{O}$: shear maps, rotation curves, BAO distance measures, covariance matrices.
- Π_T: baseline model mapping visible matter + chosen gravity model to predicted observables.
- $\mathbf{W}$: instrument/systematics covariance or likelihood geometry.
- $\mathcal{C}$: channelization into redshift bins, angular modes, spatial templates.

Domain gates:

- **Closure surrogate:** null tests (e.g., jackknife, random rotations, cross-shear nulls) must yield residual consistent with noise.
- **Identifiability:** vary nuisance models (PSF, selection, photo-z) and test stability of residual structures.
- **Intervention surrogate:** systematics perturbation replaces physical intervention.

Pass condition: residual structure persists across validated pipelines and is not eliminated by plausible systematics models.

5 Hypothesis Ladder and Model Comparison (Only After Gates)

5.1 Three-level hypothesis ladder

After Level 1 passes and at least one Level 2 domain passes, hypotheses may be compared.

Hypothesis 5.1 (H0: Baseline completeness)**.** Under the specified contract, baseline physics + systematics fully explain observables; residuals are consistent with noise.

Hypothesis 5.2 (H1: Baseline + conventional extension)**.** Residuals are reduced by adding an established extension class (e.g., additional known backgrounds, standard parameter extensions, or conventional DM/DE parameterizations).

Hypothesis 5.3 (H2: Void-ledger physical interpretation (optional))**.** If H0 and H1 fail after systematics audits, and residual structure is cross-domain stable, then a new degree of freedom (interpretable via the Void ledger) may be required.

5.2 Decision rule (evidence, not vibe)

Use likelihood/evidence-based comparisons under the contract:

- report $\Delta \log \mathcal{L}$, information criteria, or Bayesian evidence ratios,

- penalize model complexity,
- pre-register which residual structures count as "signal" vs "systematic".

Remark 5.4. The LTT is compatible with both frequentist and Bayesian workflows. The key requirement is *contract explicitness* and *gate discipline*.

6 Outputs: What "Done" Looks Like

6.1 Artifacts

A successful program yields:

1. **Tool certificate** with pass/fail gate matrix, thresholds, and limitations.
2. **Domain reports** (Lab / Detector / Cosmology), each including:
   - frozen contract,
   - calibration sources,
   - systematics audit table,
   - residual plots and structure-test outputs,
   - intervention/perturbation results.
3. **Hypothesis comparison dossier** (H0/H1/H2):
   - evidence metrics,
   - sensitivity analysis,
   - falsification conditions.

6.2 Stop conditions (when to halt interpretation)

Stop and revise the tool/contract if:

- closure fails in lab,
- identifiability fails (loss vs model error),
- residual structures vanish under small systematics perturbations,
- the tool requires post-hoc tuning to "find" structures.

A Appendix A: Gate Checklist (Printable)

Gate 1: Well-definedness

☐ $\mathcal{O}$ defined with units/calibration and uncertainty.

☐ Π_X, Π_T map into same $\mathcal{O}$.

☐ $\mathbf{W}$ valid (PSD), matches $\mathcal{O}$ ordering.

☐ $\mathcal{C}$ defined pre-analysis.

Gate 2: Closure

☐ Closed accounting simulation passes.

☐ Real lab closure passes within uncertainty.

Gate 3: Identifiability

☐ Missing-channel injection routes residual into correct channel.

☐ Systematic model bias triggers "model error" flag, not Void credit.

Gate 4: Intervention

☐ Directional predictions match under parameter change.

☐ Residual structure stable under small perturbations.

B Appendix B: Minimal Contract Template

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Field	Required content

Field	Required content

Domain Lab / Detector / Cosmology (choose one)
$\mathcal{O}$ Observable vector definition (components, units, calibration)
Π_X Matter-side projection procedure (inputs, outputs, assumptions)
Π_T Energy/field-side projection procedure (baseline model, nuisance params)
Metric $\delta = \sqrt{r^\top \mathbf{W} r}$ with $\mathbf{W} = \mathbf{\Sigma}^{-1}$ or likelihood deviance
$\mathcal{C}$ Channelization map (pre-registered)
Ledgers V_E, V_I update rules h_E, h_I
Systematics Known nuisance list + how perturbed in audits
Intervention Physical knob (lab) or perturbation protocol (cosmo)
Pass/Fail Thresholds for gate decisions